Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2017
PI Name:	Bailey, Michael R. Ph.D.		
Project Title:	A Phantom and Test Bed to Enhance Kidney Stone Comminution by Therapeutic Ultrasound		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Task Description:	Original project aims/objectives This proposal is relevant to research emphasis Sec. A.2.e Smart Medical Systems and Technology Team of NASA Research Announcement (NRA) NNJ15ZSA001N-NSBRI (National Space Biomedical Research Institute). It addresses the research topic described in the NRA: Employing a Suitable Ground Analog Environment, Develop and Test an Ultrasound Phantom or Surrogate Testing Capabilities to Enhance the Utility of Therapeutic Ultrasound. The proposed work will help mitigate the Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities and help close Gap ExMC 4.13: We have limited capability to screen for, diagnose, and treat renal stones during exploration missions as well as Gap ExMC 4.02: We do not have the capability to provide non-invasive medical imaging during exploration missions. Kidney stones are common on Earth, and space travel only exacerbates the risk. A Russian mission was nearly aborted because of a kidney stone and over 30 stone incidences associated with space travel are reported in the evidence base for Gap ExMC 4.13 medical condition Nephrolithiasis. Over 15 years of NIH (National Institutes of Health) and NSBRI funded research have developed burst wave lithotripsy (BWL), which comminutes stones faster and more completely than shock wave lithotripsy (SWL) in a bench top water bath. The University of Washington has already licensed the technology to a commercial partner, and NSBRI is funding BWL development for integration into NASA ExMC's Flexible Ultrasound (FUS) system. However, neither SWL nor BWL comminutes stones as well in a confined space, such as the kidney, and both perform even less well when the stone is obstructing and surrounded by tissue. This is arguably because surface cavitation critical to each process is inhibited by the confinement. In addition, cavitation is acutely sensitive to carbon dioxide, which is 30 times more soluble than oxygen in water, and carbon dioxide levels can be elevated 20 times in NASA vehicles. As such, this proposal discovered effective BWL exposures to counteract the effects of confinement and carbon dioxide. Aim 1 was to test the effect of confinement on BWL in a phantom. Aim 2 was to measure the pressure threshold for BWL comminution in an analog environment. Key findings: We developed and validated a series of phantoms replicating the ureter, kidney, and human torso. This model system was used to optimize outputs to successfully comminute human stones in these tissue mimicking spaces. The model system provides a training and development tool for future development and application. Our findings measure the effect of water temperature, gas content, volume of enclosure, stone composition, stone size, tissue attenuation, and other factors on comminution efficiency of both SWL and BWL. We used the findings to create a phantom and protocol to most accurately simulate conditions and comminution in the human body. We use that phantom to complete the first comparison of SWL and BWL. We demonstrated human stones could be comminuted with BWL within the confined collecting space of a phantom or porcine model and in a porcine model breathing elevated carbon dioxide. Thus, successful setting for BWL have been validated in a space analog environment. We discovered the optimal outputs combine the outputs of BWL with ultrasonic propulsion (UP) developed with NSBRI to reposition stones. The BWL pulses fracture the stone and the UP pulses separate the collection of fragments. Our probes were modified to produce both the UP and BWL pulses. These probes function with the BWL amplifier made to integrate with the NASA FUS and to produce both UP and BWL pulses. These data, probes, and outputs have been integrated into a commercial COTS device developed by industry and NSBRI for integration within the NASA FUS to add the capability to fragment stones in space. Although elevated CO2 has been shown to affect the capability to detect and image stones, our works shows burst wave lithotripsy remains capable of fragmenting stones under elevated CO2 conditions possible in NASA vehicles. Our study provides data to improve understanding of the mechanisms of action of BWL and SWL. These and other data support BWL is less reliant on cavitation to comminute stones than is SWL. Our study also provides information on the presence of bubble nuclei within the body and rate of exchange of gasses from the blood to the urine. Bubbles appear to be present on the stones and the gasses within these bubbles appear to change within minutes of breathing elevated CO2. These bubbles can be detected by twinkling artifact on ultrasound and if detected in the blood could potentiate decompression sickness in the right conditions. Cavitation can also cause tissue injury in therapeutic ultrasound and breathing elevated CO2 significantly lowered the cavitation threshold in the bladder of one of our pigs. This reinforces the value of testing therapeutic ultrasound in a flight analog. Impact: In the language of the NRA NNJ15ZSA001N-NSBRI directive, with the above Aims, we developed and validated an ultrasound phantom and surrogate testing capabilities in the ground analog environment of an elevated carbon dioxide atmosphere to enhance the utility of therapeutic ultrasound for kidney stone comminution. The work provides an option to manage a mission critical stone episode. Proposed research plan for the coming year: A study is planned to test a training protocol and proficiency of emergency medicine physicians. We are pursuing adding the BWL pulses to our current FDA-approved trial of UP in the UW (University of Washington) Emergency Department to provide the first in man trial of BWL and to test the first definitive management of acute stones. BWL is being integrated into the FUS as a portable painless noninvasive method to fragment stones in space. BWL and UP along with advanced stone imaging are also being commercialized for Earth use.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Kidney stones have long been near the top of NASA's list of concerns; mitigating Gap 4.02 medical condition Nephrolithiasis is a shall for all missions beyond the International Space Station (ISS). Likewise, stones have plagued humans since ancient Egypt. Currently, one in eleven Americans has suffered from stones -- more than have diabetes or cardiovascular disease. Dehydration, stasis, and bone demineralization are strong contributors to kidney stones, and occur in microgravity, increasing the risk of stones in space. Stones are often debilitating, and pilots cannot fly with stones. Stones occurred on a Russian space mission, and the mission was nearly aborted before the stone passed. Over 30 stones have occurred shortly following even short duration space flights. NASA has collected compelling evidence for concern on its website (Ed. note: site not available at this time; 11/9/2017). Additionally, since the website publication, the total number of astronaut stone episodes has more than doubled, and a drug introduced to combat visual impairment/intracranial pressure has exacerbated the risk. Science, experience, and the negative medical consequences support concern for the risk of stones in space. NASA and NSBRI have focused considerable attention on stones and made progress. However, there are many types of stone disease, and it is unlikely that stone disease will ever be completely prevented on Earth or in space. We propose a way to prevent or minimize the consequences of any stones that form while in space.The treatment for most kidney stones is to encourage natural passage. To quote NASA's expectations in space Based on current Lifetime Surveillance of Astronaut Health (LSAH) data, 80 to 85% of in-flight cases of nephrolithiasis are expected to be best case scenarios (defined as a renal stone that responds to conservative treatment, e.g., analgesics and hydration), and 15 to 20% would be worst case scenarios (defined as a renal stone that does not respond to conservative treatment, e.g., requires lithotripsy or surgical treatment). Even surgery leaves residual fragments that must pass. Our technology provides the capability to image, fragment, and reposition stones within the

	kidney and ureter, which will enhance conservative treatment or surgery by accelerating and facilitating passage of stones or fragments. However, this does not have to be the only use. The technology can also be used to reposition a stone to a non-obstructing location within the kidney to postpone surgery or to accelerate passage through the ureter, as proposed here. In addition, this grant adds to the growing body of knowledge of how elevated carbon dioxide in space vehicles affects physiology and space medicine.
Task Progress:	A Russian mission was nearly aborted because of a kidney stone and over 30 stone incidences associated with space travel are reported in the evidence base for Gap ExMC 4.13 medical condition Nephrolithiasis. NIH and NSBRI funded research have developed Burst Wave Lithotripsy (BWL), which comminutes stones faster and more completely than Shock Wave Lithotripsy (SWL) in a water bath. NSBRI is funding BWL development for integration into NASA ExMC's Flexible Ultrasound system. However, neither SWL nor BWL comminutes stones as well in a confined space, such as the kidney, arguably because surface cavitation critical to each process is inhibited by the confinement. Cavitation is also acutely sensitive to CO2, and CO2 levels can be elevated 20x in NASA vehicles. This proposal seeks effective BWL exposures to counteract the effects of confinement and CO2. Aim 1: Test effect of confinement on BWL in a phantom. Hypothesis: The pressure threshold and exposure duration to comminute stones will be higher in the confined space of the kidney or uretero-pelvic junction (UPJ) compared to a free field. Rationale: Preliminary studies have shown the close proximity of tissue-mimicking material to the stone can delay BWL treatment. The confinement of the UPJ is greater than that of the renal pelvis, and stones commonly obstruct at the UPJ and become symptomatic. The outcome was to define BWL outputs to effectively comminute stones in confined spaces simulating the urinary collecting space. Aim 2: Measure Pressure threshold for BWL. Comminution in an analog environment. Hypothesis: The pressure threshold to comminute stones and for concomitant injury will be higher for pigs with elevated CO2 levels. Rationale: Evidence supports that Doppler based detection of stones is caused by small surface bubbles and this capability to detect stones disappears in minutes in pigs breathing CO2 levels possible in NASA vehicles. The outcome was to define BWL outputs to effectively comminute stones in vivo in an elevated CO2 environment. These Aims were successfully completed. A mannequin phantom was developed for these tests and as a continued training and research tool. Outputs were developed and measured to effectively comminute human kidney stones in the confinement of the kidney and ureter phantoms and in animal models breathing elevated carbon dioxide. In the language of the NRA NNJ15ZSA001N-NSBRI directive, with the above Aims, we develop[ed] and test[ed] an ultrasound phantom and surrogate testing capabilities in the ground analog environment of an elevated CO2 atmosphere to enhance the utility of therapeutic ultrasound for kidney stone comminution. These validated outputs have been added to the BWL system integrated with NASA's flexible ultrasound system (FUS).
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